TIMOTHY SINGH

Software Engineer / Solutions Architect

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SUMMARY

Software Engineer with an M.Sc. in Data Science from UBC (GPA: 4.33) and a B.Sc. in Computer Science (First Class Honours). Experienced building Python packages, REST/API-enabled applications, web systems, Java/JDBC database interfaces, and Linux-based technical solutions across government, financial services, and applied research settings. Brings hands-on experience with Python, Java, JavaScript, Flask, Node.js, Express.js, SQL, Docker, GitHub Actions, AWS, Ubuntu, and ROS/Gazebo. Known for writing maintainable code, validating systems before deployment, and translating technical requirements into reliable software used by real stakeholders.

EDUCATION

M.Sc. Data Science — ***GPA: 4.33*** | *2025*
University of British Columbia, Vancouver, Canada

B.Sc. Computer Science - First Class Honours — ***GPA: 4.23*** | *2021*
University of the West Indies, St. Augustine Campus

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, C#, SQL, Bash, PHP, Dart, XSLT
Backend & APIs: Flask, Node.js, Express.js, Java Spring Boot, ASP.NET, REST APIs, JDBC, SQLAlchemy
Frontend & Web: HTML5, CSS3, JavaScript, Angular, Progressive Web Apps, responsive design, XML, XSLT
Databases: MySQL, PostgreSQL, Oracle, SQLite, database design, CRUD operations, PHPMyAdmin
Cloud, DevOps & Tools: AWS (S3, EC2, Lambda), Docker, GitHub Actions CI/CD, Git, Heroku, Apache, Postman
Systems & Infrastructure: Ubuntu 20.04/22.04, Ubuntu Server, Proxmox VE, LDAP / Active Directory, Bash scripting, network imaging
Testing & Quality: Bug testing, responsiveness testing, system validation, code review, edge-case analysis, technical documentation
Specialized & ML Software: ROS, Gazebo, PyTorch workflow integration, scikit-learn, model deployment and monitoring

EXPERIENCE

AI Content Development Assistant | Ministry of Education — *April 2026 – Present*

- Designed and built an agentic AI video and audio generation pipeline, using Python scripts and API requests to synchronize generated media while leveraging AWS for processing and storage.
- Managed a production GitHub repository by overseeing team permissions, maintaining GitHub Actions CI/CD workflows, and establishing a structured development workflow to support collaboration across the developer team.

Solutions Architect (Capstone) | ALS Geoanalytics — *April - July 2025*

- Improved UI/UX for a carbon emissions dashboard tracking regional emissions across Canadian provinces, making emissions trends and workflow impacts easier to interpret for technical and non-technical users.
- Containerized development and testing with Docker and maintained automated CI/CD workflows using GitHub Actions for reproducible package delivery.
- Built an end-to-end Python pipeline integrating specialized Python workflows, PyTorch training loops, and real-time carbon intensity APIs to automate hourly, emissions-aware measurement with minimal codebase changes.
- Designed, developed, and benchmarked a Python package for measuring carbon emissions from large-scale deep learning workflows, achieving 85% accuracy compared to existing packages.
- Implemented tracking for optimization techniques including pruning and quantization, measuring up to a 20% reduction in model training carbon footprint.

AI Data Reviewer | DataAnnotation — *July 2025 - Present*

- Evaluate code, reasoning, and text-generation outputs for correctness, efficiency, instruction adherence, and edge-case handling using structured scoring rubrics.
- Deliver technical feedback on model outputs, directly supporting software-quality improvement and iterative model refinement.

Solutions Architect | Ministry of Digital Transformation *Sept 2021 - Aug 2024*

- Developed and deployed the Ministry website using HTML, CSS, JavaScript, PHP, XML, and XSLT while collaborating across teams in agile workflows.
- Conducted bug testing, responsiveness testing, and system validation for Ministry digital initiatives, improving reliability and usability before deployment.
- Implemented and maintained an open-source asset management system with Active Directory integration, improving tracking accuracy and deployment efficiency by 35%.
- Configured, deployed, and maintained Ubuntu 20.04 environments for 200+ staff using Bash scripting and network imaging, reducing onboarding time by 40%.
- Engineered ICT Access Centre technology deployments across 8 communities, delivering hardware and open-source software solutions to support 1,500+ residents.
- Provided Tier 0-3 technical support across hardware, software, and network issues, strengthening troubleshooting skills across production environments.

Software Engineer | Republic Bank - ITMD *June - Aug 2020*

- Designed and implemented a secure Java/JDBC front-end interface in Apache NetBeans, allowing administrators to manage database records without direct backend access.
- Supported a user recertification and database clean-up project, improving record accuracy and reducing administrative overhead for the Production Support - Credit Card team.
- Cleaned and transformed large Excel datasets using Python and pandas for ingestion into an internal SQL database, improving accessibility for analytics users.

Support Associate | Republic Bank *Oct 2017 – July 2018*

KEY PROJECTS

Project Bloodline *(2020)*
Developed a Progressive Web Application that encouraged blood donation through a user-friendly interface. Built with the JAM stack using Angular 9.1.1 on the front end and Flask 1.0.2 on the back-end application server, then deployed on Heroku.
Website: project-bloodline.web.app

dsresumatch *(2025)*
Open-source Python package that extracts resume and job description sections, structures unstructured text, and scores keyword relevance. Published with automated testing via GitHub Actions CI/CD.
GitHub: github.com/UBC-MDS/dsresumatch

CariVoice *(2026)*
An LLM that primarily utilizes speech to turn into text and product images. The primary use case is to be used as a transcription system for telling Caribbean folklore stories and plays, while being supplemented with images.
GitHub: github.com/OAOverflowAnalytics/CariVoice

Robotic Arm Automation - Virtana Collaboration *(2021)*
Programmed a Mecademic500 robotic arm for precise pick-and-place operations using ROS and Gazebo in a Linux environment. Designed and tested robotic automation workflows through simulation and motion control.
Demo: youtu.be/JDTCNuB09mc

Credit Card Default Predictor *(2025)*
End-to-end Python project using XGBoost, LightGBM, and RandomForest on roughly 30,000 records. Built reproducible ingestion, feature engineering, model evaluation, and CI/CD workflows.
GitHub: github.com/SimplyTim/Credit-Card-Default-Predictor

CERTIFICATIONS

- Create Machine Learning Models in Microsoft Azure (2024)
- IBM - Generative AI: Introduction and Applications (2024)
- IBM - Generative AI: Prompt Engineering Basics (2024)
- Project Management Fundamentals & Applications (2023)